Brightlearn Data Analytics
Exercise 4: SQL Foundations: Union

01) SELECT account_id,
account_holder,
'Sandton' AS city
FROM branch_sandton_accounts
UNION
SELECT account_id,
account_holder,
'Rosebank' AS city
FROM branch_rosebank_accounts;

02) SELECT product_code,
product_name,
'Savings' AS product_type
FROM savings_products
UNION
SELECT product_code,
product_name,
'Current' AS product_type
FROM current_products;

03) SELECT staff_id,
staff_name,
NULL AS email
FROM retail_banking_staff
UNION
SELECT staff_id,
staff_name,
NULL AS email
FROM corporate_banking_staff;

04) SELECT city_code,
city_name,
'Mobile' AS region
FROM mobile_branch_cities
UNION
SELECT city_code,
city_name,
'digital' AS region
FROM digital_branch_cities;

05) SELECT customer_id,
customer_name,
'Targeted' AS segment
FROM push_notification_targets
UNION
SELECT customer_id,
customer_name,
~~'TARGET~~ 'Targeted' AS segment
FROM inapp_banner_targets;

06) SELECT transaction_id,
account_id,
amount,
NULL AS transaction_date
FROM atm01_transactions
UNION ALL
SELECT transaction_id,
account_id,
amount,
NULL AS transaction_date
FROM atm02_transactions

07) SELECT application_id,
customer_id,
loan_type,
NULL AS amount requested
FROM gauteng_loan_application
UNION ALL
SELECT application_id,
customer_id,
loan_type,
NULL AS amount requested
FROM western_cape_loan_application ;

08) SELECT complaint_id,
customer_id,
'App' AS category,
NULL AS logged_date
FROM app_complaints
UNION ALL
SELECT complaint_id,
customer_id,
'Email' AS category
NULL AS logged date
FROM email_complaints ;

04) SELECT payment_id,
account_id,
amount,
'april' AS payment_date
FROM april_payments
UNION ALL
SELECT payment_id,
account_id,
amount
'may' AS payment_date
FROM may_payments ;

10) SELECT entry_id,
account_id,
entry_type,
NULL AS amount,
NULL AS entry_date
FROM debit_entries
UNION ALL
SELECT entry_id,
account_id,
entry_type,
NULL AS amount,
NULL AS entry_date
FROM credit_entries ;

Bonus Questions

01) The fraud analyst must use UNION, as he or she wants a unique list of the customers that appear in the high-risk watch list whether it is for the January or February one.
UNION removes duplicate rows, so customers who appears on both list would only be included once.

02) The auditing must use UNION ALL, because they need every transaction record to be included in the total. UNION ALL includes all rows and don't remove duplicates, this would ensure that row count includes transactions recorded in both systems

03) Both SELECT statements in a UNION must return the same numbe of columns and the corresponding columns must have compatible data.
The query failed because the first SELECT returns only 2 columns, whereas the 2nd SELECT statement returns 3 columns
For it ~~the~~ to work, the query should be as follows

```
SELECT  account_holder,
        account_id
FROM    branch_sandton_accounts
UNION
SELECT  account_holder,
        account_id
FRUM    branch_rosebank_accounts ;
```